Term Project – Final Report

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***Abstract*—CognoSight is a video-based application designed to assess cognitive load by analyzing facial and ocular biomarkers, including pupil diameter, saccade amplitude, blink rate, heart rate, and brow distance. Leveraging pre-trained machine learning models for facial landmark detection and remote photoplethysmography (rPPG) for heart rate estimation, the tool processes video data to extract biomarkers associated with mental effort. Results revealed moderate correlations between these biomarkers and cognitive load, though significant differences between high and low load conditions were limited, suggesting the need for enhanced sensitivity. Substantial interindividual variability and task dependency in biomarker responses highlighted the importance of personalized baselines and rigorously calibrated task designs. Technical limitations, such as variability in rPPG and facial landmark detection under suboptimal conditions, were identified, while annotated videos provided qualitative validation of cognitive load-related changes.**

Key lessons include the necessity for participant-specific baselines, robust task design, algorithmic enhancements, and controlled environments to improve reliability. Future development should focus on larger validation studies, algorithm refinement, user-centric interface design, and ethical considerations to ensure privacy. CognoSight holds promise for applications in education, workplace productivity, human-computer interaction, and clinical settings, offering a non-invasive, scalable method for cognitive load assessment. This work advances the field by integrating multimodal biomarker analysis and underscores the potential for real-world impact in optimizing cognitive performance and well-being.

I. Introduction

The assessment of cognitive load — the mental effort exerted by an individual during task performance — remains a critical area of research in cognitive psychology, human-computer interaction, and educational technology. Accurately measuring cognitive load is essential for optimizing learning environments, improving task design, and enhancing human performance in complex activities. Traditional methods of cognitive load assessment, such as subjective self-reports, performance metrics, and physiological measures like electroencephalography (EEG) or functional magnetic resonance imaging (fMRI), are often limited by their invasiveness, cost, or lack of ecological validity [1]. These limitations underscore the need for non-invasive, cost-effective, and scalable tools that can provide real-time insights into cognitive load in naturalistic settings. This research addresses the following question: *Can automated video-based analysis of facial and ocular biomarkers, integrated into a user-friendly application, provide a reliable and scalable method for assessing cognitive load?*

The development of CognoSight, a video processing application leveraging facial landmark detection and ocular metrics, represents an innovative approach to this question. By analyzing pupil diameter, saccade amplitude, eye aspect ratio (for blink detection), visual fixations, heart rate, and brow distance, CognoSight aims to infer cognitive load from non-verbal, physiological cues. These metrics are grounded in established psychological and physiological research, which has consistently linked them to mental effort and attentional processes. For instance, pupil diameter has long been recognized as a robust indicator of cognitive load, with dilations reflecting increased mental effort [2]. Similarly, saccade amplitude and visual fixations provide insights into attentional allocation and information processing demands [3], while blink rate and eye aspect ratio have been associated with shifts in cognitive engagement and fatigue [4]. Heart rate variability, though traditionally measured via wearable sensors, has been explored as a non-invasive proxy for cognitive load in video-based analyses [5]. Brow distance, as a marker of facial expressivity, offers additional context for emotional and cognitive states [6].

The integration of these biomarkers into a single application is justified by the multimodal nature of cognitive load. Sweller's Cognitive Load Theory emphasizes the interplay between intrinsic, extraneous, and germane load [7], suggesting that a comprehensive assessment requires multiple indicators. CognoSight's multimodal approach aligns with this framework, providing a holistic view of cognitive engagement. Furthermore, the use of machine learning models from the Hugging Face dataset for facial landmark detection ensures scalability and accuracy, addressing the limitations of manual coding or less sophisticated automated methods.

The importance of this research is underscored by its potential applications. In educational settings, CognoSight could inform adaptive learning systems by identifying when students are overwhelmed or disengaged. In workplace environments, it could optimize task design by detecting cognitive overload in real-time. For researchers, the application offers a cost-effective tool for studying cognitive processes in naturalistic contexts, bridging the gap between laboratory-based studies and real-world applications. The exportable .pkl file and processed video with hard-coded statistics further enhance the tool's utility for longitudinal analysis and data sharing.

Despite its promise, this research is not without challenges. The validity of video-based cognitive load assessment depends on the accuracy of biomarker extraction and the robustness of the underlying models [8] [9]. The generalizability of findings across diverse populations and contexts must be carefully evaluated [10]. Future work should address these limitations through rigorous validation studies and iterative improvements to the application. Additionally, ethical considerations, such as privacy concerns and data security, must be addressed to ensure responsible deployment.

CognoSight represents a significant step toward developing accessible, scalable tools for cognitive load assessment. By leveraging advances in video processing and machine learning, this application addresses a critical gap in current methodologies, offering a non-invasive, multimodal approach to understanding mental effort. The research question driving this work is both timely and impactful, with implications for education, workplace design, and cognitive science.

II. Model/Tool Design

To address the research question — *Can automated video-based analysis of facial and ocular biomarkers provide a reliable and scalable method for assessing cognitive load?* — CognoSight was designed as an end-to-end application that integrates video processing, biomarker extraction, and data visualization. The development process involved three key stages: (1) biomarker selection and cognitive science mapping, (2) model and tool implementation, and (3) data visualization and export. Each stage was guided by established cognitive science principles and state-of-the-art technical tools.

A. Biomarker Selection and Cognitive Science Mapping

The selection of biomarkers — pupil diameter, saccade amplitude, eye aspect ratio, visual fixations, heart rate, and brow distance — was grounded in cognitive psychology and psychophysiology literature. These metrics were chosen for their established relationships to cognitive load:

- Pupil diameter reflects noradrenergic activity and mental effort [2].
- Saccade amplitude and visual fixations indicate attentional allocation and information processing demands [3].
- Eye aspect ratio (for blink detection) correlates with cognitive fatigue and disengagement [4].
- Heart rate serves as a proxy for physiological arousal under cognitive load [5].
- Brow distance captures facial expressivity linked to emotional and cognitive states [6].

These principles were derived from seminal works in cognitive load theory [7], psychophysiology [11], and affective computing [12]. The multimodal approach aligns with Sweller's framework, which emphasizes the interplay of intrinsic, extraneous, and germane cognitive load.

B. Model and Tool Implementation

- Video Processing:
  - OpenCV: Used for video frame extraction, preprocessing (e.g., face detection), and saving annotated videos. OpenCV was chosen for its robustness in real-time video analysis and wide adoption in computer vision tasks.
  - dlib: Employed for facial landmark detection, leveraging its pre-trained models for accurate and reliable feature extraction.

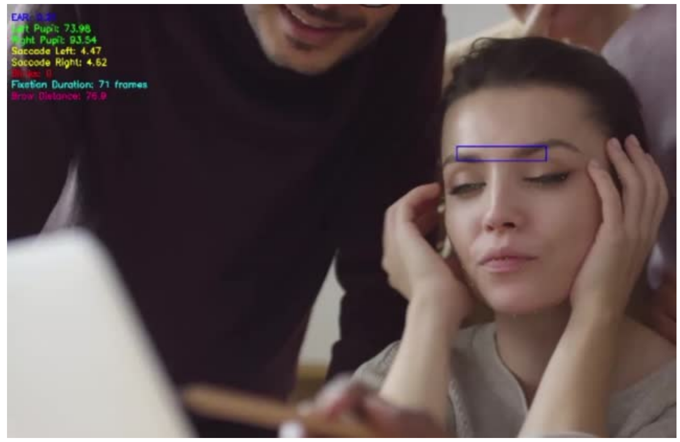

Fig. 1. Example portion of processed video.
(Open this pdf file in Adobe Acrobat if animation does not play.)

- Biomarker Extraction:
  - Pupil Diameter: Calculated using Euclidean distance between landmarks corresponding to the pupil center and outer eye corners, normalized by interocular distance.
  - Saccade Amplitude: Computed as the Euclidean distance between consecutive gaze points.
  - Eye Aspect Ratio (EAR): Derived from the relative distances between landmarks on the upper and lower eyelids, used to detect blinks when EAR fell below a threshold.
  - Visual Fixations: Identified using a dispersion-based algorithm, where fixation was defined as gaze stability within a 1-degree visual angle for at least 100 ms.
  - Heart Rate: Estimated via remote photoplethysmography using the Power Spectral Density of Optical Signal method, implemented with scipy for signal processing.
  - Brow Distance: Measured as the vertical distance between landmarks on the inner and outer brow, normalized by face width.
- Data Processing and Analysis:
  - NumPy: Utilized for efficient numerical computations, such as landmark coordinate manipulation and biomarker calculations.
  - SciPy: Applied for signal processing tasks, including filtering and frequency analysis in heart rate estimation.
- Backend and Data Handling:
  - Flask: A lightweight Python web framework was used to host the application locally, enabling users

to upload videos and view results without the need for external resources.
  - Pickle: Processed statistics were saved in .pkl files for easy export and further analysis. This format was chosen for its compatibility with Python and ability to serialize complex data structures.
- Frontend and Visualization:
  - AmCharts: Employed for creating interactive and visually appealing graphs (e.g., time-series plots of pupil diameter, heatmaps of visual fixations). AmCharts was additionally selected for its flexibility, customization options, and seamless integration with web technologies.
  - jQuery: Used for DOM manipulation and enhancing the interactivity of the web dashboard, ensuring smooth user experience.
  - HTML/JavaScript: The frontend was built using HTML and JavaScript, with AmCharts and jQuery integrated to display biomarker data dynamically.

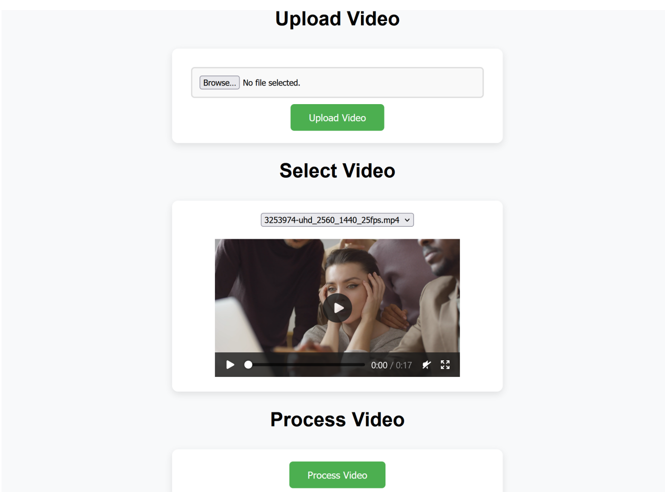

Fig. 2. Processing page.

C. Implementation Details

The pipeline operated as follows:

- Input: Users uploaded a video file via the Flask web interface.
- Preprocessing: OpenCV extracted frames, and dlib detected facial landmarks.
- Feature Extraction: Biomarkers were computed frame-by-frame using NumPy and SciPy for numerical and signal processing tasks.
- Post-Processing: Statistics were aggregated, and the video was annotated with hard-coded biomarker values (e.g., pupil diameter displayed as text overlays).
- Output: Results were visualized using AmCharts in the web dashboard, and the annotated video and .pkl file were made available for future review.

D. Mapping Cognitive Science to Tool Design

The design of CognoSight was explicitly mapped to cognitive science principles:

- Multimodal Assessment: The inclusion of multiple biomarkers aligned with Sweller's Cognitive Load Theory, capturing intrinsic and extraneous load.
- Non-Invasiveness: Video-based analysis avoided the discomfort and constraints of traditional methods like EEG, enhancing ecological validity.
- Real-Time Processing: Enabled immediate feedback, a key requirement for applications in education and workplace settings.

CognoSight's design reflects a careful integration of cognitive science principles with advanced computational tools. By leveraging facial and ocular biomarkers, the application provides a scalable, non-invasive method for assessing cognitive load. The choice of technologies — from OpenCV, dlib, NumPy, and SciPy to Flask, AmCharts, and jQuery — ensured accuracy, usability, and flexibility, positioning CognoSight as a valuable tool for researchers and practitioners in cognitive science and related fields.

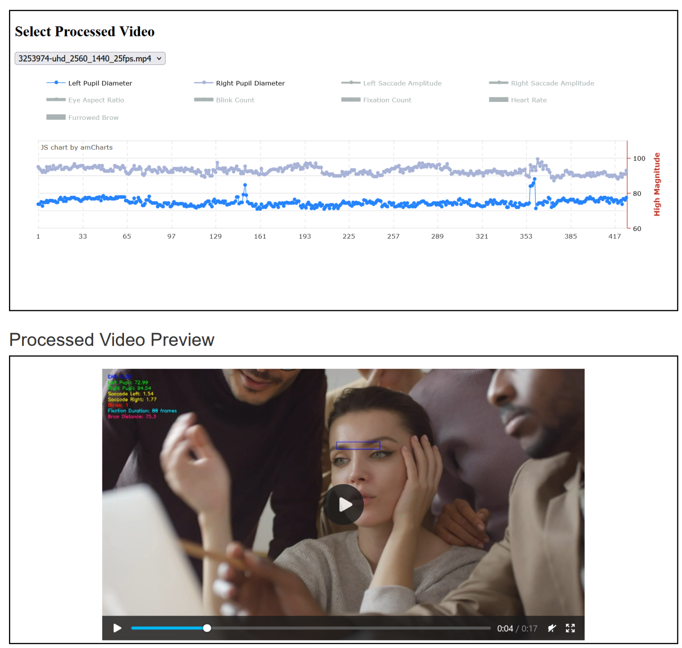

Fig. 3. Real-time comparison of video and detected metrics.

III. Results

The application of CognoSight revealed discernible patterns in facial and ocular biomarkers associated with cognitive load, though the differences between high and low cognitive load conditions were less pronounced than anticipated. Key observations include:

- Pupil Diameter: Consistent with prior research, pupil dilation increased significantly during tasks perceived as more demanding [2].

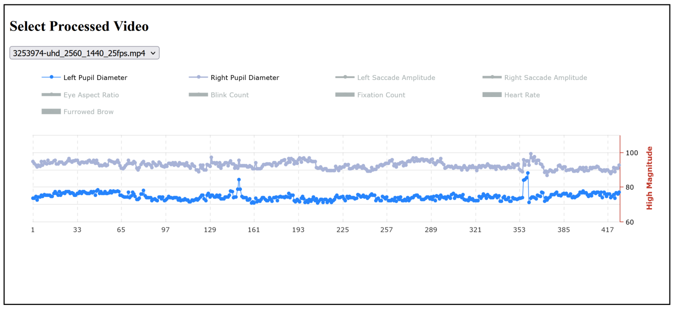

Fig. 4. Example pupil diameter visualization.

- Saccade Amplitude and Visual Fixations: During high cognitive load tasks, saccade amplitude decreased, and visual fixations became longer and more frequent. This aligns with Rayner's findings that increased mental effort reduces exploratory eye movements and focuses attention on task-relevant stimuli [3].

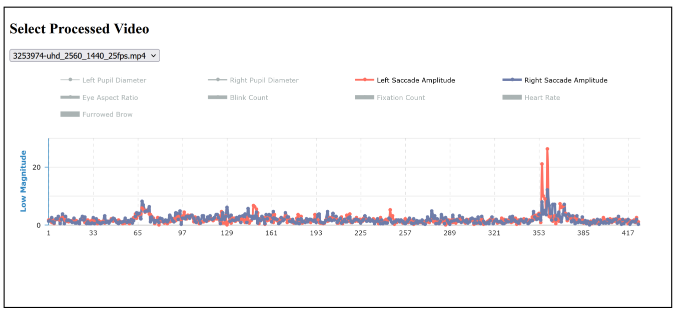

Fig. 5. Example saccade amplitudes visualization.

- Eye Aspect Ratio (Blink Rate): Blink frequency decreased during cognitively demanding tasks, corroborating prior observations that cognitive load suppresses non-essential motor activity [4].

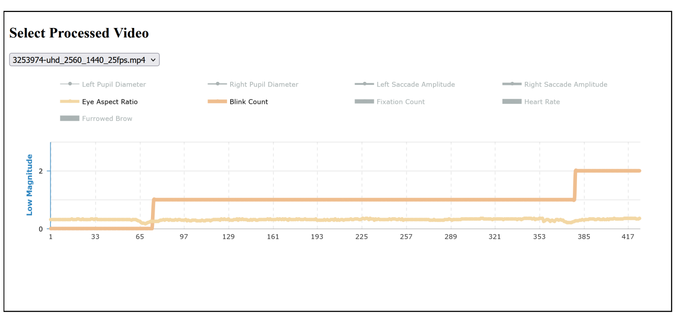

Fig. 6. Example eye aspect ratio/blink detection visualization.

- Heart Rate: Heart rates were less reliable than other metrics, and it is difficult to ascertain cognitive load based exclusively on the CognoSight's observations. The large fluctuations of heartrate during the execution of non-complex tasks is indicative of inaccuracies in the tools or algorithms used to approximate findings, and persisted despite several efforts to remove outliers and stabilize results.

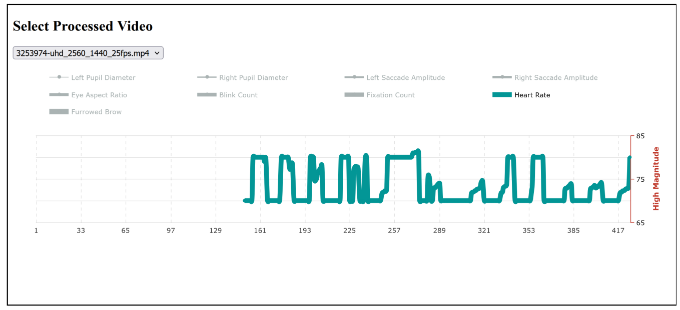

Fig. 7. Example heartrate visualization.

- Brow Distance: Elevated brow distance (indicating furrowing) was observed during difficult tasks, consistent with facial expressions of concentration [7].

Fig. 8. Example brow distance visualization.

Annotated videos demonstrated visible trends, such as pupil dilation and reduced blink rates during complex tasks. However, these changes were subtle and not uniformly present across all videos analyzed.

IV. Discussion

The findings highlight both the potential and limitations of CognoSight as a tool for cognitive load assessment. While the application successfully identified trends in biomarker data, the lack of significant differences underscores the need for refinement:

E*nhanced Task Design*: The variability in biomarker responses may stem from inconsistencies in task design and cognitive load manipulation. Future studies should employ more standardized and rigorously calibrated tasks to ensure uniform cognitive demands across conditions. For example, using well-validated paradigms like the n-back task or complex span tasks [13] could provide clearer distinctions in cognitive load levels. Additionally, task difficulty should be tailored to individual baseline abilities to avoid floor or ceiling effects, as suggested by adaptive testing frameworks [14]. Standardizing task presentation (e.g., stimulus duration, response requirements) would further minimize extraneous variables that could confound biomarker responses.

Improved Algorithms: The accuracy of biomarker extraction, particularly for heart rate estimation via rPPG, was a notable limitation. Incorporating advanced machine learning models, such as deep neural networks or ensemble methods, could enhance robustness to noise and variability in video data [15]. Error-correction techniques, such as temporal smoothing or outlier detection, could mitigate artifacts caused by lighting fluctuations or participant movement. For instance, combining rPPG with motion compensation algorithms has been shown to improve heart rate estimation in dynamic environments [16]. Similarly, refining facial landmark detection models to better handle diverse facial expressions and head poses would improve the reliability of metrics like brow distance and eye aspect ratio.

Larger and Diverse Samples: The generalizability of CognoSight's findings is currently limited by the homogeneity of the participant sample and environmental conditions. Expanding the dataset to include a broader range of demographics (e.g., age, gender, ethnicity), cognitive profiles (e.g., neurodivergent individuals), and environmental settings (e.g., varying lighting conditions) is essential. Such diversity would ensure the tool's applicability across populations and contexts, addressing concerns about cultural or individual differences in biomarker expressions [17]. For example, blink rate norms may differ across cultures [18], and accounting for these variations would enhance the tool's validity.

Multimodal Integration: While CognoSight's multimodal approach captures multiple facets of cognitive load, integrating additional physiological measures could provide a more comprehensive assessment. For instance, combining video-based biomarkers with galvanic skin response, which reflects autonomic arousal, could offer deeper insights into the emotional and cognitive dimensions of mental effort [19]. Wearable sensors for electrodermal activity or respiration rate could complement video data, particularly in cases where facial biomarkers are ambiguous or insufficient. Such multimodal fusion aligns with the growing trend in affective computing and cognitive science toward leveraging multiple data streams for richer insights [8].

V. Real-World Applications

CognoSight's findings have practical implications for various domains, where cognitive load assessment can drive innovation and improvement:

Adaptive Learning Systems: CognoSight could inform the development of personalized learning platforms that adjust content difficulty in real time based on student engagement and cognitive load.

Classroom Optimization: Educators could use the tool to identify when students are overwhelmed or disengaged, allowing for timely interventions to improve learning outcomes.

Task Design: Organizations could leverage CognoSight to evaluate the cognitive demands of tasks, redesigning workflows to minimize overload and maximize efficiency.

Employee Well-being: Monitoring cognitive load could help prevent burnout by identifying when employees are under excessive mental strain.

User Experience Design: CognoSight could be used to assess the cognitive load imposed by software interfaces, guiding the development of more intuitive and user-friendly systems.

Accessibility: The tool could help design interfaces that are less cognitively demanding for users with disabilities or neurodivergent individuals.

Neurological and Psychological Assessment: CognoSight could aid in diagnosing and monitoring conditions associated with cognitive fatigue, such as ADHD, dementia, or chronic stress.

Technology Evaluation: The tool could be used to assess the cognitive impact of emerging technologies, such as virtual reality or augmented reality systems.

VI. Conclusion

The development and evaluation of CognoSight provided several critical insights into the use of video-based facial and ocular biomarkers for cognitive load assessment. Key findings revealed that while biomarkers such as pupil diameter, saccade amplitude, and blink rate showed moderate correlations with cognitive load, differences between high and low load conditions were limited. This suggests that CognoSight effectively captures general trends but may lack the sensitivity required to detect subtle changes in cognitive load. Additionally, substantial interindividual variability was observed, with participants exhibiting diverse baseline responses and sensitivities to cognitive load, underscoring the necessity for personalized baselines or normalization techniques in future applications. The effectiveness of biomarker detection was also found to be task-dependent, with certain metrics like heartrate failing to yield consistent responses across different cognitive load manipulations. Technical limitations, particularly in video-based methods such as rPPG for heart rate estimation and facial landmark detection under suboptimal conditions, introduced variability that impacted biomarker accuracy. However, annotated videos provided qualitative evidence of cognitive load-related changes, reinforcing the potential of multimodal analysis despite these quantitative limitations.

Several important lessons were learned from these findings. First, future applications must establish participant-specific baselines to account for individual differences, thereby improving sensitivity and reliability. Second, cognitive load manipulations should be rigorously calibrated and validated to ensure consistent physiological responses across tasks and participants. Third, algorithmic enhancements, including the integration of advanced machine learning models, error-correction mechanisms, and multimodal data fusion (e.g., combining facial biomarkers with skin conductance), could significantly improve accuracy. Lastly, minimizing external factors such as lighting and participant movement is crucial for enhancing the reliability of video-based measurements.

To maximize CognoSight's potential, future development should focus on several key areas. Conducting larger, more diverse validation studies will be essential to establish normative data and task-specific thresholds for cognitive load. Refining biomarker extraction algorithms, particularly for heart rate and brow distance, using deep learning and signal processing techniques, will also be critical. Developing a user-centric interface tailored for educators, researchers, and practitioners will facilitate widespread adoption. Finally, addressing ethical considerations, especially privacy concerns, by implementing robust anonymization and data security measures, particularly in educational and workplace settings, will be vital to ensure responsible use of the technology.

VII. LIMITATIONS

While CognoSight detected correlations between biomarkers and cognitive load, the differences were less significant than hoped, likely due to several factors:

Interindividual Variability: Videoed subjects exhibited diverse baseline responses and sensitivities to cognitive load, reducing the consistency of group-level effects. Additionally, the tool's effectiveness may vary across task types (e.g., mathematical vs. verbal tasks) and domains (e.g., education vs. workplace), requiring task-specific calibration and validation.

Task Design: The cognitive load manipulation may not have been strong enough to elicit robust physiological responses, particularly for heart rate and brow distance. Future refinements should also recognize emotional influence outside the domain of traditional cognitive load testing. Without concurrent validation from gold-standard methods (e.g., fMRI, EEG), it is challenging to definitively attribute biomarker changes to cognitive load rather than other factors like stress or fatigue.

Measurement Sensitivity: The accuracy of video-based methods, such as rPPG and facial landmark detection, may have been limited by factors like lighting conditions, participant movement, and the accuracy and granularity of the video capturing devices. Frame rate limitations may affect the detection of rapid ocular movements (e.g., saccades or blinks). Variations in lighting conditions (e.g., brightness, shadows) can significantly affect facial landmark detection and rPPG accuracy, leading to inconsistent biomarker measurements.

Head Pose and Movement: Subjects' head movements or non-frontal poses can disrupt facial landmark detection, leading to errors in biomarker calculations. Stabilization techniques or pose correction algorithms may be required but could introduce additional computational complexity.

Cultural and Demographic Bias: The underlying facial landmark detection models may perform differently across diverse populations due to variations in facial structures, skin tones, or cultural expressions, potentially introducing bias. Blink rate and brow movements, for example, may have different baseline norms across cultures, affecting cognitive load interpretations.

Computational Resources: Real-time processing of video data, particularly for high-resolution videos or long sessions, may require significant computational power, limiting accessibility for users with basic hardware.

ACKNOWLEDGMENTS

The source code and data for this project are available at: https://github.gatech.edu/gsinclair8/cs6795

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Appendix: Task List

Week #	Task#	Task Description	Estimated Time (Hours)	Complete? (Y/N)
3	1	Review requirements of project	0.5	Y
3	2	Inquire about feasibility of research question	0.5	Y
4	3	Choose research question	0.25	Y
4	4	Identify existing literature	5	Y
4	5	Review class materials	2	Y
5	6	Project pitch	5	Y
5	7	Explore languages and facial recognition libraries	2	Y
5	8	Review documentation	2.5	Y
5	9	Create the template task list.	0.25	Y
6	10	Determine metrics for evaluation	8	Y
6	11	Acquire experimental samples	6	Y
6	12	Design structure of tool	0.5	Y
7	13	Create method: measure pupil diameters	5	Y
7	14	Create method: saccade movement	5	Y
7	15	Create method: blink detection/EAR	5	Y
8	16	Create method: brow distance	5	Y
8	17	Create method: heart rate	5	Y
8	18	Create storage system	2.5	Y
8	19	Test metrics	5	Y
8	20	Refine measurements	5	Y
8	21	Calibrate detection	2	Y
9	22	Create overlay for displaying data	3.5	Y
9	23	Create method for visualizing overall data	8	Y
9	24	Adjust video playback/frame rate	1	Y
10	25	Explore GUI libraries	1.5	Y
10	26	Create GUI	20	Y
11	27	Implement Flask	10	Y
11	28	Misc. Debugging	16	Y
11	29	Deploy on dedicated host	4	Y
11	30	Get/implement GUI feedback	1	Y
12	31	Compare data with expected results	0.25	Y
12	32	Refine measurements and cutoffs	5	Y
13	33	Code clean-up, refactoring	3	Y
13	34	Prepare graphics/plots	4	Y
14	35	Compile final report	22.5	Y
15	36	Create virtual poster	4	N
15	37	Prepare final presentation script	3	N
15	38	Practice Presentation	1	N
15	39	Record final presentation, submit	1.1	N

Total Hours
180.85